

RL for Bilateral Price Negotiation

Final Project Team

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Roadmap

- 1 Problem Formulation
- 2 Data and SFT
- 3 GRPO Method
- 4 Training Curriculum
- 5 Results and Demo
- 6 Lessons

Task: Private-Value Bargaining

- Build a second-hand marketplace negotiation agent with one base LLM.
- The same model plays two roles through system prompts and adapters:
 - Buyer: close a deal below the private maximum budget.
 - Seller: close a deal above the private minimum acceptable cost.
- The model must bargain naturally while obeying strict action formats.

Why RL?

SFT can teach language and format, but final bargaining quality depends on interactive outcomes: deal price, legality, leakage, and walkaway decisions.

Environment and Action Grammar

- State: item, market reference, hidden role value, and dialogue history.
- Terminal outcomes: legal deal, role violation, walkaway, timeout, or format error.
- Difficulty comes from the bargaining zone:

$$gap = \frac{budget - cost}{cost}$$

Parser Targets

```
[PRICE: 1220]  
<deal>1220</deal>  
<walkaway>
```

Private Information

The agent should not reveal budget, cost, floor price, or ceiling price.

Scenario Data: Controlled Bargaining Difficulty

- Scenarios are rule-generated from product templates and generation profiles.
- RL pool: 4000 train, 500 validation, 500 test scenarios.
- The pool intentionally emphasizes hard negotiations.

Split	Total	Near-zero	Narrow	Balanced	Wide
Train	4000	1008	1426	980	586
Validation	500	114	163	139	84
Test	500	118	193	124	65

Key point from the project plan: scenarios are RL inputs; GRPO rollouts are generated online during training.

SFT Teaches the Negotiation Language

- SFT is not optimized for final economic reward.
- It teaches role identity, multi-turn rhythm, valid action syntax, and deal/walkaway behavior.
- Data pipeline:

```
scenario templates  
→ scenario sampler  
→ API-generated dialogues  
→ turn-level SFT messages  
→ buyer/seller training splits
```

Design Choice

Thinking-style SFT examples are written into message content rather than a separate field, matching the training format consumed by LLaMA-Factory.

GRPO Uses Online Self-Play

- 1 Sample a scenario from the RL scenario pool.
- 2 Let the current buyer and seller policies produce a complete dialogue.
- 3 Parse the terminal outcome and compute role-specific reward.
- 4 Normalize rewards within sampled groups to obtain GRPO advantages.
- 5 Update only the currently active role adapter.

Why not reuse SFT dialogues?

Fixed dialogues do not expose the policy to its own mistakes. Online rollouts make reward depend on the current interaction behavior.

Two Adapters, Two Views of One Trajectory

Runtime Policies

base + buyer adapter
base + seller adapter

- Same base model.
- Separate LoRA adapters.
- Only active role tokens receive loss.

Avoided Failure Mode

Do not mix buyer and seller rewards on the same completion; their objectives are intentionally opposed.

Trajectory View

Seller update: history may include buyer turns, completion is seller turn, reward is seller utility.

Buyer update: history may include seller turns, completion is buyer turn, reward is buyer utility.

Reward: Utility plus Safety Constraints

$$u_{\text{buyer}} = \frac{\text{budget} - \text{price}}{\text{budget} - \text{cost}}$$
$$u_{\text{seller}} = \frac{\text{price} - \text{cost}}{\text{budget} - \text{cost}}$$

- Legal deals reward each role by normalized surplus.
- Invalid deals below seller cost or above buyer budget receive strong penalties.
- Shaping terms penalize format errors, leakage, extreme offers, and bad monotonicity.
- Balanced V1 adds a shared Nash-style term:

$$\text{shared} = \sqrt{u_{\text{buyer}} \cdot u_{\text{seller}}}$$

Three Stages from the Training Plan

Stage	Updated adapter	Purpose
1	Buyer only	Learn to bargain down, respect budget, and walk away when needed.
2	Seller only	Learn anchoring, concession, and protection above cost.
3	Alternating	Let both sides adapt, with lower learning rate and KL control.

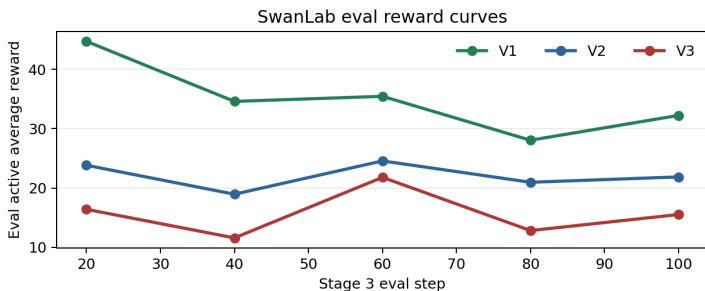
Engineering Rule

When Stage 3 alternates roles, gradient accumulation must be tracked per adapter, otherwise optimizer steps can mix buyer and seller updates.

Engineering Iterations and Evidence

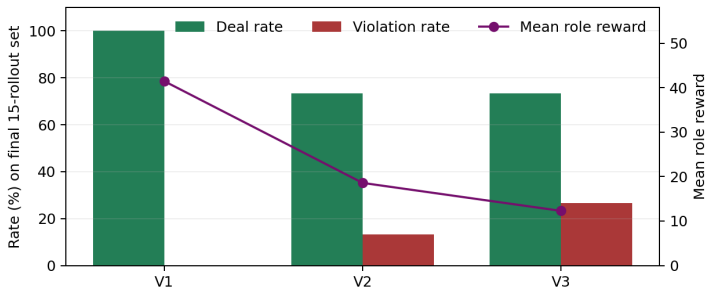
- Persisted eval results locally to compare checkpoints beyond SwanLab screenshots.
- Hardened rewards so buyers no longer profit from below-cost seller violations.
- Added a cold-seller guard in Stage 1 to prevent exploitation of a weak opponent.
- Compared Stage 2 seller checkpoints; step_100 preserved high legal-deal rate while giving buyer some surplus.
- Added split buyer/seller reward metrics to avoid misleading alternating curves.

Training Reward Curves from SwanLab



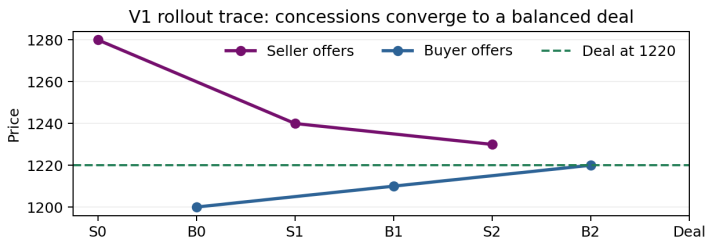
V1 has the strongest eval reward curve among the three Stage 3 variants. V2/V3 add more behavior guardrails, but the reward signal becomes harder to optimize and the final policy is less stable.

V1 / V2 / V3 Ablation: Final Policies



Final rollout comparison shows the trade-off clearly: V1 gives the best deal rate and reward; V2/V3 guardrails reduce some qualitative risks but introduce more failed or violating outcomes.

Trace: A Balanced V1 Rollout



Scenario: AirPods Pro 2. Buyer budget is 1412, seller cost is 1034, and the deal closes at 1220. The surplus split is buyer 50.8% and seller 49.2%.

Trace Dialogue: V1 Rollout Index 47

Seller	1280: condition is close to new; this is already a meaningful concession.
Buyer	1200: I am serious and can pay immediately if this works.
Seller	1240: I can move down another step, but not much further.
Buyer	1210: I can add 10 and close now.
Seller	1230: final small concession.
Buyer	1220: one more 10; if accepted I will buy now.
Seller	<deal>1220</deal>

This example was selected because both roles receive positive and nearly equal reward: buyer 56.9 and seller 56.0.

What Worked, What Failed, What We Learned

Worked

- SFT gave stable language and syntax.
- Online GRPO changed bargaining behavior.
- Balanced V1 produced a usable economic demo.

Failed / Limited

- Early Stage 3 became seller-dominant.
- V1 still had leakage and label residue in some samples.
- V2/V3 guardrails improved intent but hurt violation or deal metrics.

Takeaways

- The central contribution is not only the model checkpoint, but the environment and training loop that turn negotiation into measurable RL.
- Role-specific adapters are necessary because buyer and seller rewards are opposed even when the dialogue is shared.
- Reward design determines the bargaining equilibrium; stronger penalties are not automatically better.
- Next step: add human-rated dialogue quality, explicit leakage classifiers, and more diverse opponents.